

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Project Synopsis on

“Design and Implementation of Low-Power 32-bit RISC-V Processor for IOT and Biomedical Applications”

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATION ENGINEERING

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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Certificate

This is to certify that following students

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Has submitted Project synopsis on **“Design and Implementation of Low-Power 32-bit RISC-V Processor for IOT and Biomedical Applications”** for VI Semester B.E. in Electronics & Communication Engineering during the academic year 2025-26. The project synopsis has been approved as it satisfies the academic requirements in respect of Project work prescribed for the Bachelor of Engineering Degree.

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Design and Implementation of Low-Power 32-bit RISC-V Processor for IOT and Biomedical Applications

Introduction

Modern biomedical devices such as wearable health monitors and implantable sensors are used for continuous monitoring of physiological signals. These devices measure important signals such as Electrocardiography (ECG) and Electroencephalography (EEG), which help in diagnosing and monitoring various health conditions. Continuous monitoring allows doctors to observe a patient's health status in real time and provide timely medical support.

Since most biomedical devices operate on small batteries, the processors used in these systems must be designed to consume very low power while maintaining reliable performance. Low-power processor design is important to extend battery life, reduce heat generation, and ensure safe operation in portable and implantable devices. Therefore, energy-efficient embedded processors play a key role in the development of modern biomedical systems.

RISC-V is an open-source and customizable processor architecture that provides flexibility for designing application-specific processors. Its simple and modular design allows developers to optimize the processor for low power consumption and efficient performance, making it suitable for biomedical and IoT applications.

This project focuses on designing a low-power, resource-efficient RISC-V processor optimized for biomedical applications in portable healthcare devices. The processor includes key components such as instruction memory, register file, arithmetic logic unit (ALU), control unit, and data memory to process biomedical data efficiently while minimizing hardware resources and power consumption.

Problem Statement

1. Conventional processors like arm cortex m series often consume higher power (ranges from 10mW to 500mW) and require larger hardware resources, making them inefficient for compact biomedical systems.
2. There is a need for a customizable [user friendly, flexible] and energy-efficient processor architecture that can handle biomedical signal processing with minimal power consumption
3. Low-Power Resource-Efficient RISC-V Processor for Biomedical Applications (approximately power consumption is from 10mW to 50mW)

Objectives of the Proposed Project

1. To design a 32-bit low-power processor based on the RISC-V suitable for biomedical and IoT applications. The design focuses on achieving efficient processing while minimizing power consumption for portable and embedded healthcare devices.
2. To implement the processor architecture using Cadence tools and verify its functionality through simulation. This ensures correct instruction execution and proper operation of all processor modules.
3. To evaluate the performance of the designed processor by analysing parameters such as power consumption, processing speed, and hardware utilization. This analysis helps determine the efficiency and suitability of the processor for low-power biomedical systems.
4. To optimize the processor architecture in order to reduce power consumption and improve overall efficiency. The optimization aims to enhance performance while maintaining a compact and resource-efficient processor design.

Literature Survey

[1] Sharma, V. P., Kumar, A., Khan, M. F., Lokhande, M., & Vishvakarma, S. K. (2025) proposed Bio-RV, a low-power and resource-efficient processor based on the RISC-V architecture for biomedical applications. The processor is designed to process physiological signals such as ECG and EEG in portable healthcare devices. The architecture focuses on reducing power consumption while maintaining sufficient computational capability for continuous health monitoring. By optimizing hardware resources and simplifying the processor architecture, the design becomes suitable for wearable and implantable medical devices that require reliable and energy-efficient operation.

[2] Kurali, S., & Fidan, H. presented the implementation of a 32-bit RISC-V processor on FPGA to demonstrate the practical hardware realization of the architecture. Their work shows how a processor designed using hardware description languages can be synthesized and deployed on programmable hardware platforms. The study evaluates parameters such as execution performance, hardware utilization, and system functionality. The results highlight the feasibility of implementing RISC-V processors in embedded systems and provide insights into efficient hardware resource usage.

[3] Yang, Y., et al. (2023) proposed a low-power IoT RISC-V processor integrated with a hybrid encryption accelerator to enhance security in connected devices. Since IoT systems frequently transmit sensitive data, the processor includes hardware acceleration for encryption to improve security without significantly increasing power consumption. The design demonstrates how additional functional modules can be integrated with the RISC-V architecture while maintaining energy efficiency. This approach is particularly useful for IoT devices that require secure data transmission and low-power operation.

[4] Narla, S., et al. (2025) presented a study on the complete RISC-V processor design flow from RTL to GDSII using electronic design automation tools such as Cadence. Their work explains the different stages involved in digital integrated circuit design, including architecture design, RTL coding, synthesis, place and route, and final layout generation.

The research demonstrates how a full processor chip can be developed using standard VLSI design methodologies and verified through simulation and analysis.

[5] Teyene, A., & Taconit, J. proposed a low-power RISC-V processor core optimized for edge computing devices. Their design focuses on minimizing energy consumption while maintaining sufficient processing capability for edge-based applications such as IoT sensing and data processing. The architecture emphasizes lightweight design techniques and efficient instruction execution to reduce overall power usage. This work highlights the importance of energy-efficient processor design for devices that operate with limited power resources.

[6] Park, J., et al. (2024) developed a low-power RISC-V multicore processor with a shared floating-point unit (FPU) for IoT end-node systems. In this architecture, multiple processor cores share a common floating-point unit, which helps reduce hardware area and power consumption. The shared design improves computational efficiency while minimizing redundant hardware resources. The proposed architecture demonstrates how multicore processing can be implemented efficiently in resource-constrained IoT devices.

These research works demonstrate significant progress in the development of energy-efficient processors using the RISC-V architecture. They highlight various approaches such as hardware optimization, FPGA implementation, security integration, and multicore processing to improve performance and efficiency. Inspired by these studies, the present work focuses on designing a low-power 32-bit RISC-V processor optimized for biomedical and IoT applications, with emphasis on reducing power consumption and improving overall processing efficiency.

Proposed Methodology

The development of the proposed 32-bit low-power RISC-V processor follows a structured design flow. Initially, the processor architecture is designed with modules such as Instruction Memory, Control Unit, ALU, Register File, and Data Memory. The design is then simulated using Verilog and HDL testbenches to verify functionality. After

analysing power, speed, and hardware utilization, the architecture is optimized before implementation and testing on FPGA.

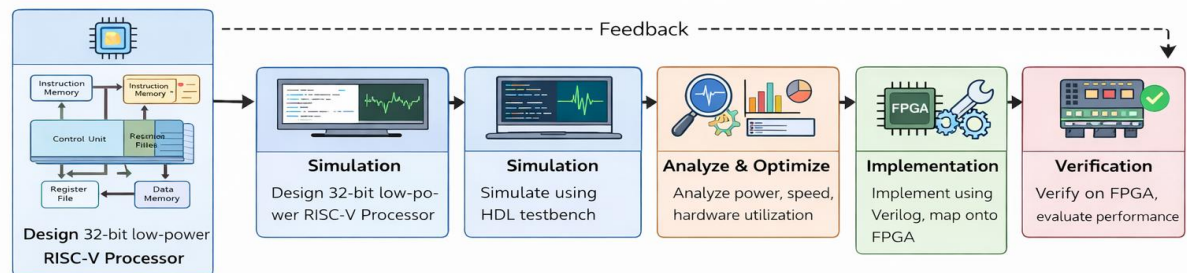


Figure -1: Block Diagram

1. Design Stage: The processor architecture is designed with components such as Instruction Memory, Control Unit, ALU, Register File, and Data Memory, focusing on low power and efficient processing.
2. Simulation: The design is simulated using Verilog and HDL testbenches to verify the correct operation of each module.
3. Analyse and Optimise: The processors are analysed for power, speed, and hardware utilization, and the architecture is optimized to improve efficiency.
4. Implementation: The optimized design is implemented using Verilog and synthesized using EDA tools, then mapped onto an FPGA platform.
5. Verification/Testing on FPGA: Finally, the processor is tested on FPGA to verify functionality and evaluate real-time performance.

Issues and Challenges

The major design challenges include:

1. Maintaining very low power consumption for continuous biomedical device operation.
2. Designing the processor with minimal hardware resources and chip area.

3. Ensuring high reliability and safety for critical medical applications.
4. Handling real-time processing requirements in medical monitoring systems.
5. Optimizing the implementation of the RISC-V instruction set architecture for biomedical applications.
6. Performing extensive testing and verification to ensure correct processor functionality.
7. Balancing low power consumption with adequate processing performance.

Expected Outcomes

The expected outcomes of this project are:

1. Reduced power consumption [approx. to 10mW to 50mW] in biomedical signal processing systems.
2. Efficient hardware utilization with a compact processor design.
3. Improved performance using the RISC-V architecture.
4. Enhanced suitability for wearable and portable biomedical devices.

Applications

The proposed sensor can be used in:

1. Used in smart wearables that continuously monitor vital signals such as heart rate, ECG, and body temperature.
2. Suitable for devices implanted inside the body, such as pacemakers, where low power consumption is critical.
3. Used in portable medical equipment to process physiological signals like ECG or EEG efficiently.
4. Enables continuous monitoring of patient health data and transmission to doctors or healthcare cent

Conclusion

The proposed biomedical processor design focuses on developing a low-power and resource-efficient processing system suitable for modern medical applications. By utilizing the RISC-V instruction set architecture, the design benefits from an open, flexible, and customizable platform that enables efficient implementation for specialized biomedical systems. The processor is designed to achieve reduced hardware complexity, low energy consumption, and reliable operation, which are essential requirements for implantable and wearable medical devices.

Overall, the implementation supports efficient biomedical signal processing and coordination of medical modules, making it suitable for applications such as wearable health monitoring, implantable medical devices, and remote patient monitoring systems. The proposed approach contributes to the development of compact, reliable, and energy-efficient biomedical electronic systems.

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